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EE1060: Differential Equations and Transform Techniques

Lecture 20: Fourier Series

Topics :

1. Vector Space of Functions and Orthogonality
 2. Fourier series coefficients
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Inner Product, Norm and Orthogonality of Functions (Vector space of functions)

Inner product: $\phi(x) \in V$ and $\psi(x) \in V$.

$$\langle \phi(x), \psi(x) \rangle \text{ on } I = [a, b] = \int_a^b \phi(x) \psi(x) dx \quad \text{as long as } \phi(x): \mathbb{R} \rightarrow \mathbb{R} \text{ \& } \psi(x): \mathbb{R} \rightarrow \mathbb{R}$$

for Complex valued functions $\langle \phi(x), \psi(x) \rangle = \int_a^b \phi(x) \psi(x)^* dx$.

$$\langle \phi(x), \psi(x) \rangle = \langle \psi(x), \phi(x) \rangle^*$$

Norm of a function :-
(L_2 Norm)

$$\|\phi(x)\|_2 = \sqrt{\langle \phi(x), \phi(x) \rangle} = \sqrt{\int_a^b |\phi(x)|^2 dx}$$

↑
Complex valued.

if $\phi(x)$ is real valued signal, $\|\phi(x)\|_2^2 = \int_{-\infty}^{\infty} |\phi(x)|^2 dx \rightarrow \text{energy of the signal.}$
on $[-\infty, \infty]$

orthogonality :- $\phi(x)$ and $\psi(x)$ are orthogonal on $[a, b]$ if

$$\langle \phi(x), \psi(x) \rangle = 0 \text{ on } [a, b]$$

Ex: $\cos x$ and $\cos 2x$.

Fourier Synthesis as Linear combination of Orthogonal Functions

Let $\{\phi_1(x), \dots, \phi_n(x)\}$ denote a set of functions.

$$\sum_{i=1}^n c_i \phi_i(x) \leftarrow \text{linear combination}$$

Fourier Series:- $\{\phi_n(x)\}_{n=-\infty}^{\infty}$ and $\phi_n(x) = e^{jn\omega_0 x}$ (Set of orthogonal functions)

Synthesis Equation: $g(x) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 x} \rightarrow \text{linear Comb of } \{\phi_n(x)\}_{n=-\infty}^{\infty}$

c_n : linear Comb = $g(x)$. (Target)

$g(x) \rightarrow$ Periodic with Period 'L'

$$\omega_0 = \frac{2\pi}{L}$$

Property 1:- $\phi_m(x)$ and $\phi_k(x)$ are orthogonal $\forall m \neq k$. $I \in [0, L]$

$$\langle \phi_m(x), \phi_k(x) \rangle = \int_0^L e^{jm\omega_0 x} \cdot e^{-jk\omega_0 x} dx = 0. \quad I \in [a, a+L]$$

Property 2:- $\|\phi_k(x)\| = \sqrt{L}$. $m, I \in [0, L]$

Determination of Fourier Coefficients and Simplification for Real valued functions

$$g(x) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 x} \rightarrow \textcircled{1}$$

$g(x) \rightarrow$ periodic with period L

$$\omega_0 = \frac{2\pi}{L}$$

if $\textcircled{1}$ holds, can we compute c_k ?

Take inner product w.r.t $\phi_k(x) \Rightarrow \langle g(x), \phi_k(x) \rangle = \sum_{n=-\infty}^{\infty} \langle c_n \phi_n(x), \phi_k(x) \rangle$

$$= c_k \underbrace{\|\phi_k(x)\|^2}_{=L}$$

$$\Rightarrow c_k = \frac{\langle g(x), \phi_k(x) \rangle}{\langle \phi_k(x), \phi_k(x) \rangle} = \frac{\langle g(x), \phi_k(x) \rangle}{\underbrace{\|\phi_k(x)\|^2}_{=L}}$$

$$c_k = \frac{1}{L} \int_0^L g(x) e^{-jk\omega_0 x} dx$$

← Fourier Coefficient
Fourier Analysis Eqn

↓
Projection of $g(x)$ on $\phi_k(x)$

When $g(x)$ is real, $c_{-k} = ?$

$$c_{-k} = \frac{1}{L} \int_0^L g(x) e^{-j(-k)\omega_0 x} dx = \left(\frac{1}{L} \int_0^L g(x) e^{-jk\omega_0 x} dx \right)^* = c_k^*$$

$c_{-k} = c_k^*$

Projection of function onto Orthogonal Set

Let $\{\phi_1(x), \dots, \phi_n(x)\}$ be an orthogonal set on I .

functions. Let $g(x)$ be defined in I .

determine $c_1, \dots, c_n$ such that

$$D = \|g(x) - \underbrace{\sum_{i=1}^n c_i \phi_i(x)}_{=}\|^2 \text{ is minimum}$$

Result :- The set of $c_1, \dots, c_n$ that achieves a unique minimum

$$c_i = \frac{\langle g(x), \phi_i(x) \rangle}{\langle \phi_i(x), \phi_i(x) \rangle} \rightarrow \textcircled{2}$$

$$f(x) = \sum_{i=1}^n c_i \phi_i(x) \rightarrow \textcircled{3}$$

↓ ↘ = (2)

Projection onto $\{\phi_1, \dots, \phi_n(x)\}$

from linear algebra:-

$\text{Proj}_{\vec{b}} \vec{a}$ (Projection of $\vec{a}$ on to $\vec{b}$) =

$$\begin{aligned} \text{(vector projection)} &: (\vec{a} \cdot \vec{e}_b) \vec{e}_b \\ &= \frac{\vec{a} \cdot \vec{b}}{\|\vec{b}\|^2} \vec{b} \end{aligned}$$

$$\text{Proj}_{\vec{w}} \vec{a} = \sum_{i=1}^m \text{Proj}_{\vec{b}_i} \vec{a}$$

$\vec{w} = \text{span}(\vec{b}_1, \dots, \vec{b}_m)$ ↘ orthogonal

$$\text{Proj}_{\vec{b}_i} \vec{a} = (\vec{a} \cdot \vec{e}_{b_i}) \vec{e}_{b_i}$$

Relation between Complex Fourier Coefficients and Trigonometric Fourier Coefficients

$$g(x) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 x}$$

$$= c_0 + c_1 e^{j\omega_0 x} + c_{-1} e^{-j\omega_0 x} + c_2 e^{2j\omega_0 x} + c_{-2} e^{-2j\omega_0 x} + \dots$$

$$= c_0 + \underbrace{(c_1 + c_{-1})}_{a_1} \cos \omega_0 x + \underbrace{j(c_1 - c_{-1})}_{b_1} \sin \omega_0 x + \dots$$

$$a_0 = c_0$$

$$a_k = c_k + c_{-k} \quad (k \neq 0)$$

$$b_k = j(c_k - c_{-k})$$